

Erdős Problem #837

EP-837

L1 - TRACTABLE

OPEN

Graph Theory

ORIGINAL PROBLEM STATEMENT

Let $k \geq 2$ and $A_k \subseteq [0, 1]$ be the set of α such that there exists some $\beta(\alpha) > \alpha$ with the property that, if $G_1, G_2, \dots$ is a sequence of k -uniform hypergraphs with $\liminf \frac{e(G_n)}{\binom{|G_n|}{k}} > \alpha$ then there exist subgraphs $H_n \subseteq G_n$ such that $|H_n| \rightarrow \infty$ and $\liminf \frac{e(H_n)}{\binom{|H_n|}{k}} > \beta$, and further that this property does not necessarily hold if $> \alpha$ is replaced by $\geq \alpha$.

What is A_3 ?

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Archive: <https://www.unsolvedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>